



**SAVEETHA INSTITUTE OF MEDICAL AND
TECHNICAL SCIENCES**



Regular Expressions in Natural Language Processing

Technical Presentation

DSA0302 – NATURAL LANGUAGE PROCESSING

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Introduction to Regular Expressions

What are Regular Expressions?

- Regular Expressions (Regex) are sequences of characters used to search, match, and manipulate text.
- They help computers recognize patterns in text efficiently.
- Regex is widely used in NLP for text preprocessing and information extraction.

NLP Applications

-  **Email Extraction** – Identify email addresses from text.
-  **Phone Number Detection** – Extract or validate phone numbers.
-  **URL Extraction** – Detect website links.
-  **Text Cleaning** – Remove unwanted characters and symbols.
-  **Pattern Matching** – Search for words, dates, IDs, and keywords.
-  **Chatbots & Search Engines** – Process user queries efficiently.

Common Regular Expression Patterns

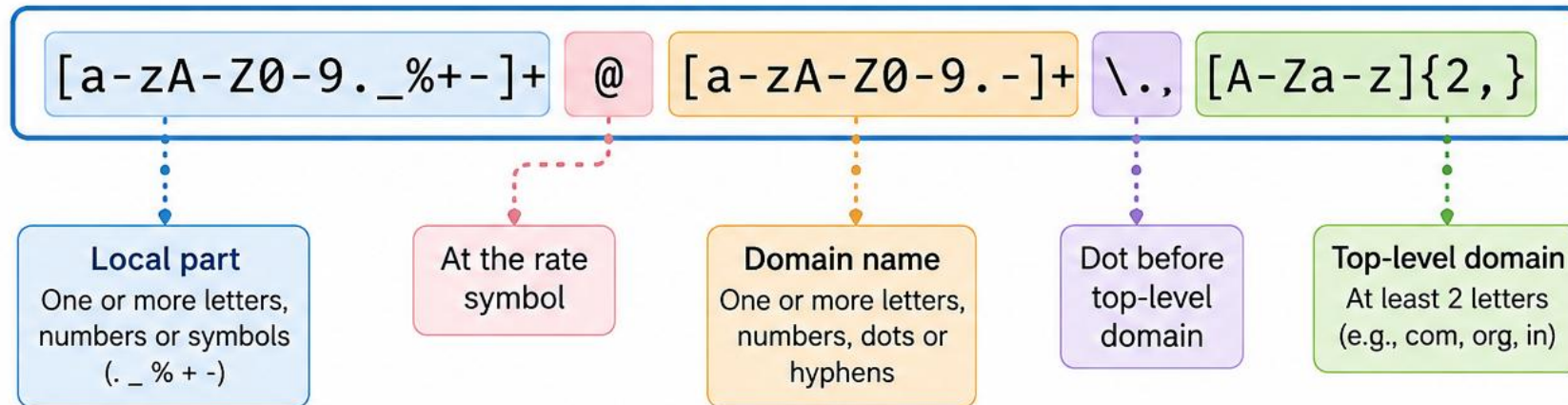
Pattern	Meaning	Example
.	Any character	a.c
*	Zero or more	ab*
+	One or more	ab+
?	Optional	colou?r
\d	Digit	123
\s	White space	space
^	Starts with	^Hello

Regex Email Matching Example

Text: Please contact us at **john123@gmail.com** for more information.

Matched Email

Regex Pattern:



Python Program

```
regex_email.py x
regex_email.py > ...
1  import re
2
3  text = "Contact us at support@example.com or admin@test.org for help."
4  pattern = r"[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}"
5  matches = re.findall(pattern, text)
6  print("Matched Emails:", matches)
7  for email in matches:
8      print("-", email)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL Python + ▢ 🗑 ^ ×

```
PS C:\NLP\RegexDemo> python regex_email.py
Matched Emails: ['support@example.com', 'admin@test.org']
- support@example.com
- admin@test.org
PS C:\NLP\RegexDemo> █
```



Advantages & Limitations



Advantages

- Fast pattern matching
- Easy to implement
- Low memory usage
- Highly efficient
- Useful in text preprocessing

Limitations

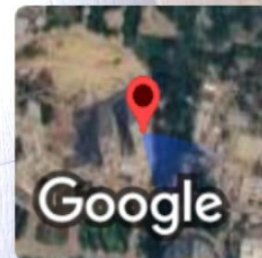
- Cannot understand sentence meaning
- Ineffective for complex grammar
- Difficult for nested structures
- Not suitable for deep semantic analysis



THANK YOU



Common Regular Expression Patterns		
Pattern	Meaning	Example
.	Any character	at
*	Zero or more	abt*
+	One or more	ab+
?	Optional	ab?c
{n}	Exact	123
{n,m}	Between values	12345
{n,}	Starts with	12345



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